

RESEARCH ARTICLE

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Adaptive control for nonlinear cyber-physical systems under false data injection attacks through sensor networks

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Funding information

National Natural Science Foundation of China, Grant/Award Number: 61621004 and 61420106016; Research Fund of State Key Laboratory of Synthetical Automation for Process Industries, Grant/Award Number: 2018ZCX03

Summary

This paper considers the adaptive control problem for a class of nonlinear cyber-physical systems with unknown nonlinearities and false data injection attacks, where the sensors are corrupted by attackers. To mitigate the effects caused by the considered attacks, a novel coordinate transformation is developed in the backstepping control design. In addition, to deal with the multiple unknown time-varying state feedback gains caused by the sensor attacks, the new types of Nussbaum functions are introduced in the adaptive control. By using Lyapunov stability theory, the proposed control scheme can guarantee all the closed-loop system signals globally bounded. Finally, the examples demonstrate the effectiveness of the proposed method.

KEYWORDS

adaptive control, false data injection attacks, globally bounded, nonlinear cyber-physical systems

1 | INTRODUCTION

Cyber-physical systems (CPSs), which are composed of physical components (actuators, sensors, and controllers) communicated with each other over a network,¹ have become common in many real applications like traffic control, safety measures and environment control, etc. Though there are many benefits of CPSs, they still face some challenges.² One is the system security problem since CPSs are connected to the internet for remote monitoring and control, which are more vulnerable to attacks.³ Thus, taking security issues into consideration is very important to ensure the safe operation of CPSs.^{4,5}

The security problems are not new to the control field, in particular in the areas of fault tolerant control⁶⁻⁸ and fault detection.^{9,10} In the control systems, controllers use sensor data for state feedback, state estimation, and output feedback. In a sensor attack, an attacker sends falsified sensor data to the controller by means of the vulnerabilities in the communication scheme, which can change the system performance and even lead to instability. Considering the sensor attacks, the control problem has received considerable attention. In the work of Yucelen et al,¹¹ the linear system with state-independent and state-dependent sensor attacks was investigated. Taking both the sensor and actuator attacks into consideration, an adaptive control strategy was developed in Jin et al.¹² Furthermore, an improved adaptive resilient control method which can ensure the asymptotic stability and transient performance was presented in the work of An and Yang.¹³ When the sensor and actuator attacks are frequency constrained, the reliable control problem for CPSs was studied in the work of Huang and Dong.¹⁴ Considering the attacks on communication links from controllers to actuators,

the observer-based attack-resilient control architecture for linear systems was proposed in the work of Xie and Yang.¹⁵ In addition, the state estimation problem against sensor attacks was considered in the works of Yang et al.^{16,17} However, these methods are mostly implemented for linear systems.

For the nonlinear systems, denial-of-service (DoS) attacks and deception attacks are commonly considered in CPSs security problem.¹⁸ The DoS attacks attempt to prevent the information transmission of sensor measurements and control inputs. Moreover, the deception attacks mean that the attackers tamper the transmitted information to affect the integrity of data, which includes replay attacks,¹⁹ false data injection attacks,^{20,21} etc. Recently, considerable attention has been paid to the nonlinear CPSs with attacks. Considering the network attacks between the nodes, the synchronization of complex dynamical networks was studied in the work of Wang et al.²² The nonlinear interconnected CPSs with intermittent DoS attacks were considered in the work of An and Yang.²³ Furthermore, the discrete-time stochastic nonlinear systems under deception attacks were considered in the work of Ding et al.²⁴ In another work by Ding et al.,²⁵ the observer-based control approach was adopted for discrete-time multiagent systems with lossy sensors and cyber-attacks. In addition, the extended Kalman filter was designed for the stochastic nonlinear systems in the work of Liu et al.,²⁶ where the attacks occur in the data transmission from sensors to remote filters. In the work of Zhang et al.,²⁷ the sampled-data consensus control problem for nonlinear multiagent systems with DoS attacks was studied, where the nonlinear function was assumed to satisfy the global Lipschitz condition. To our knowledge, there is no result on the adaptive control for nonlinear CPSs with general unknown nonlinearities and false data injection attacks.

Motivated by the above analysis, the adaptive control problem for nonlinear CPSs with false data injection attacks through sensor networks is investigated in this paper. A backstepping-based controller is proposed to address the time-varying and state-dependent sensor attacks. It is shown that all the closed-loop system signals are bounded. Compared with the existing results, the main contributions can be summarized as follows.

1. For the considered nonlinear CPSs, to deal with the state-dependent time-varying false data injection attacks, a novel coordinate transformation is developed in the backstepping design. The presented adaptive control scheme can effectively ensure the stability performance of the closed-loop systems.
2. To settle the problem of unknown multiple time-varying state feedback gains caused by the sensor attacks, the new types of Nussbaum functions are introduced, and the lemma where multiple terms of the proposed Nussbaum-type function are contained in the conditional inequality is established.

This paper is organized as follows. Section 2 formulates the problem and provides some preliminaries. The adaptive control method for the considered nonlinear CPSs is summarized in Section 3. The simulation results are given in Section 4. Finally, we conclude the work in Section 5.

Notation. R and R^n denote the set of all real numbers and the real n -dimensional space, respectively. If there is no confusion, $\hat{(\cdot)}$ means the estimate of $(\cdot)$, and $\tilde{(\cdot)}$ is the estimate error with $\tilde{(\cdot)} = \hat{(\cdot)} - (\cdot)$.

2 | PROBLEM FORMULATION AND PRELIMINARIES

Focus on the nonlinear CPSs under false data injection attacks through sensor networks described by

$$\begin{aligned}\dot{x}_i &= x_{i+1} + f_i(\bar{x}_i) \\ \dot{x}_n &= u + f_n(\bar{x}_n)\end{aligned}\quad (1)$$

with

$$\check{x}_i(t) = x_i(t) + \delta_s(x_i(t), t), \quad (2)$$

where $\bar{x}_i = [x_1, x_2, \dots, x_i]^T$ for $i = 1, 2, \dots, n$. $x = \bar{x}_n \in R^n$ and $u \in R$ are the states and input of the system. $\check{x}_i$ is the available compromised system state. $\delta_s(x_i(t), t)$ is the false data injection attacks. Similar to other works,¹¹⁻¹³ the attacks are assumed to be state-dependent and parameterized as $\delta_s(x_i(t), t) = \varpi(t)x_i(t)$ with unknown time-varying weight $\varpi(t)$. Besides, the nonlinear function $f_i(\bar{x}_i)$ is unknown.

Assumption 1. The weight $\varpi(t)$ satisfies $1 + \varpi(t) \neq 0$. Here, we assume $|\varpi(t)| \leq \bar{\varpi}$ and $|\dot{\varpi}(t)| \leq \bar{\dot{\varpi}}$ with unknown positive constants $\bar{\varpi}$ and $\bar{\dot{\varpi}}$.

Remark 1. From (2), we can obtain that $\check{x}_i(t) = (1 + \varpi(t))x_i(t)$. In the case $1 + \varpi(t) = 0$, the injected attacks just offset the system states, resulting the available compromised system states $\check{x}_i(t) = 0$. If $\check{x}_i(t) = 0$, it is not possible to construct $u(t)$ to recover the ideal system performance and stabilize the system. Thus, the assumption of $1 + \varpi(t) \neq 0$ is necessary. The attack model in (2) means that all states share the same weight $\varpi(t)$, which has been considered in other works.¹¹⁻¹³ In addition, in the case where the attacks cannot be parameterized, one can use a neural network universal function approximator to parameterize $\delta_s(x_i(t), t)$ on a compact subset.

Lemma 1 (See the work of Lin and Qian²⁸). *For any real-valued continuous function $a(x, y)$, there exist smooth functions $a_1(x) \geq 1$ and $a_2(y) \geq 1$ satisfying $|a(x, y)| \leq a_1(x)a_2(y)$.*

Definition 1 (See the work of Chen and Huang²⁹). If a continue function $\mathcal{N}(s): R \mapsto R$ satisfies the following equations:

$$\liminf_{k \rightarrow \infty} \frac{k - \int_0^k \mathcal{N}^-(s) ds}{\int_0^k \mathcal{N}^+(s) ds} = 0, \quad (3)$$

$$\liminf_{k \rightarrow \infty} \frac{k + \int_0^k \mathcal{N}^+(s) ds}{-\int_0^k \mathcal{N}^-(s) ds} = 0. \quad (4)$$

with the positive and negative truncated functions $\mathcal{N}^+(s)$ and $\mathcal{N}^-(s)$ (ie, $\mathcal{N}^+(s) = \max\{0, \mathcal{N}(s)\}$ and $\mathcal{N}^-(s) = \min\{0, \mathcal{N}(s)\}$), then it is named as $\mathcal{N}$ function and expressed as $\mathcal{N}(s) \in \mathcal{N}$. Obviously, the truncated functions satisfy $\mathcal{N}^+(s) \geq 0$, $\mathcal{N}^-(s) \leq 0$ and $\mathcal{N}(s) = \mathcal{N}^+(s) + \mathcal{N}^-(s)$.

Lemma 2 (See the work of Chen and Huang²⁹). *If $\mathcal{N}(s) \in \mathcal{N}$, then*

$$\limsup_{k \rightarrow \infty} \frac{1}{k} \int_0^k \mathcal{N}(s) ds = +\infty, \quad (5)$$

$$\liminf_{k \rightarrow \infty} \frac{1}{k} \int_0^k \mathcal{N}(s) ds = -\infty. \quad (6)$$

Remark 2. A Nussbaum function is a class function satisfying (5) and (6). Therefore, $\mathcal{N}$ function is a type of a Nussbaum function.

Lemma 3. *Define two smooth functions $V(\cdot)$ and $\theta(\cdot)$ on $[0, \infty)$ with $V(t) \geq 0$ and a smooth $\mathcal{N}$ function $\mathcal{N}(\cdot)$. Let the time-varying function $g_m(t) \in G = [g_m^-, g_m^+]$ with $0 \notin G$ for two constants g_m^- and g_m^+ satisfying $g_m^- g_m^+ > 0$. If*

$$\dot{V}(t) \leq -CV + D + \sum_{m=1}^n (g_m(t) \mathcal{N}(\theta_m(t)) + 1) \dot{\theta}_m(t), \quad (7)$$

$$\dot{\theta}_m(t) \geq 0, \forall t > 0 \quad (8)$$

hold for two positive constants C and D , then $V(t)$ and $\theta_m(t)$ are bounded over $[0, \infty)$.

Proof. See the Appendix. □

Remark 3. In this paper, the $\mathcal{N}$ function, not the general Nussbaum function, is introduced to deal with the unknown multiple time-varying state feedback gains caused by the sensor attacks. In the work of Chen,³⁰ it is proved that the existing Nussbaum function is not always effective in the scenarios of multivariable and/or time-varying control coefficients with unknown signs but the $\mathcal{N}$ function is successful in these scenarios.

Problem. Consider the system (1) under the false data injection attacks (2); the objective is to design an adaptive controller such that all the resulting closed-loop system signals are ensured to be bounded.

3 | CONTROLLER DESIGN

According to the definition of the considered false data injection attacks, we can obtain that

$$x_i(t) = \lambda(t)\check{x}_i(t) \quad (9)$$

with $\lambda(t) = (1 + \varpi(t))^{-1}$, where the real state variable x_i is unavailable.

According to Assumption 1, we assume that $\lambda(t)$ and $\dot{\lambda}(t)$ are bounded by $\lambda_m \leq |\lambda(t)| \leq \lambda_M$ and $|\dot{\lambda}(t)| \leq \bar{\lambda}$ with unknown positive constants λ_m , λ_M and $\bar{\lambda}$.

To design the controller under the sensor attacks, the new coordinate transformations are introduced

$$z_1 = x_1, \quad z_i = x_i - \lambda\alpha_{i-1}, \quad i = 2, 3, \dots, n, \quad (10)$$

which implies

$$\check{z}_1 = \check{x}_1, \quad \check{z}_i = \check{x}_i - \alpha_{i-1}, \quad i = 2, 3, \dots, n, \quad (11)$$

where α_{i-1} , ($i = 2, 3, \dots, n$) are smooth virtual controls to be designed. From (10) and (11), one can get that $z_i = \lambda\check{z}_i$.

Next, the controller will be designed step by step based on the backstepping technique.

Step 1: From the definition of the coordinate transformations, one has

$$\begin{aligned} \dot{z}_1 &= \dot{x}_1 = x_2 + f_1(x_1) \\ &= z_2 + \lambda\alpha_1 + f_1(x_1). \end{aligned} \quad (12)$$

By using Lemma 1, the following inequality holds:

$$|f_1(x_1)| = |f_1(\lambda\check{x}_1)| \leq \psi_1(\lambda)\phi_1(\check{x}_1) \leq \lambda_{1,1}\phi_1(\check{x}_1), \quad (13)$$

where $\lambda_{1,1} > 0$ is the unknown constant bound of $\psi_1(\lambda)$ and $\phi_1(\check{x}_1) \geq 1$ is a known smooth function.

Choose the Lyapunov function as

$$V_1 = \frac{1}{2}z_1^2 + \frac{1}{2}\tilde{\lambda}_1^2 \quad (14)$$

then

$$\begin{aligned} \dot{V}_1 &= z_1(z_2 + \lambda\alpha_1 + f_1(x_1)) + \tilde{\lambda}_1\dot{\tilde{\lambda}}_1 \\ &\leq \frac{1}{2}z_1^2 + \frac{1}{2}z_2^2 + z_1\lambda\alpha_1 + \varepsilon_1\check{z}_1^2\lambda_1\phi_1^2(\check{x}_1) + \frac{1}{4\varepsilon_1} + \tilde{\lambda}_1\dot{\tilde{\lambda}}_1, \end{aligned} \quad (15)$$

where ε_1 is a positive constant to be designed and $\lambda_1 = \lambda_M^2\lambda_{1,1}$ is a unknown constant.

Design the virtual control as

$$\alpha_1 = -k_1\check{z}_1 + \mathcal{N}(\theta_1)\Phi_1 \quad (16)$$

with

$$\Phi_1 = \varepsilon_1\check{z}_1\hat{\lambda}_1\phi_1^2(\check{x}_1), \quad \dot{\theta}_1 = \check{z}_1\Phi_1 \quad (17)$$

$$\tau_{1,1} = \omega_{1,1} - \sigma\hat{\lambda}_1, \quad \omega_{1,1} = \varepsilon_1\check{z}_1^2\phi_1^2(\check{x}_1), \quad (18)$$

where k_1 is a positive constant to be designed and $\tau_{1,1}$ is a tuning function.

Then, substituting (16)-(18) into (15), we have

$$\dot{V}_1 \leq -\bar{k}_1z_1^2 + \frac{1}{2}z_2^2 + (g_1(t)\mathcal{N}(\theta_1) + 1)\dot{\theta}_1 + \tilde{\lambda}_1\left(\dot{\tilde{\lambda}}_1 - \tau_{1,1}\right) - \sigma\tilde{\lambda}_1\hat{\lambda}_1 + \frac{1}{4\varepsilon_1} \quad (19)$$

with $\bar{k}_1 = k_1 - \frac{1}{2}$ and $g_1(t) = \lambda^2$.

Step 2: By using the coordinate transformations (10), we have

$$\begin{aligned}\dot{z}_2 &= z_3 + \lambda\alpha_2 + f_2(\bar{x}_2) - \dot{\lambda}\alpha_1 - \lambda\dot{\alpha}_1 \\ &= z_3 + \lambda\alpha_2 + f_2(\bar{x}_2) - \dot{\lambda}\alpha_1 - \lambda \left(\frac{\partial\alpha_1}{\partial\check{x}_1} \frac{\lambda\dot{x}_1 - \dot{\lambda}x_1}{\lambda^2} + \frac{\partial\alpha_1}{\partial\hat{\lambda}_1} \dot{\lambda}_1 + \frac{\partial\alpha_1}{\partial\theta_1} \dot{\theta}_1 \right).\end{aligned}\quad (20)$$

Similar to (13) and using Lemma 1, it follows that

$$|f_2(\bar{x}_2)| = |f_2(\lambda, \check{x}_2)| \leq \psi_2(\lambda)\phi_2(\check{x}_2) \leq \lambda_{2,1}\phi_2(\check{x}_2) \quad (21)$$

with known function $\phi_2(\check{x}_2) \geq 1$ and unknown constant $\lambda_{2,1}$ satisfying $\lambda_{2,1} \geq \psi_2(\lambda)$.

Consider the following Lyapunov function:

$$V_2 = V_1 + \frac{1}{2}z_2^2 + \frac{1}{2}\tilde{\lambda}_2^2 + \frac{1}{2}\tilde{\lambda}_0^2 \quad (22)$$

then

$$\begin{aligned}\dot{V}_2 &= \dot{V}_1 + \tilde{\lambda}_2\dot{\lambda}_2 + \tilde{\lambda}_0\dot{\lambda}_0 \\ &\quad + z_2 \left(z_3 + \lambda\alpha_2 + f_2(\bar{x}_2) - \dot{\lambda}\alpha_1 - \lambda \left(\frac{\partial\alpha_1}{\partial\check{x}_1} \frac{\lambda\dot{x}_1 - \dot{\lambda}x_1}{\lambda^2} + \frac{\partial\alpha_1}{\partial\hat{\lambda}_1} \dot{\lambda}_1 + \frac{\partial\alpha_1}{\partial\theta_1} \dot{\theta}_1 \right) \right) \\ &\leq \dot{V}_1 + \frac{1}{2}z_2^2 + \frac{1}{2}z_3^2 + z_2\lambda\alpha_2 - z_2\lambda \left(\frac{\partial\alpha_1}{\partial\check{x}_1} \check{x}_2 + \frac{\partial\alpha_1}{\partial\theta_1} \dot{\theta}_1 + \frac{\partial\alpha_1}{\partial\hat{\lambda}_1} \dot{\lambda}_1 \right) + \tilde{\lambda}_2\dot{\lambda}_2 + \tilde{\lambda}_0\dot{\lambda}_0 \\ &\quad + \varepsilon_1\check{x}_2^2\lambda_0 \left(\alpha_1 - \frac{\partial\alpha_1}{\partial\check{x}_1} \check{x}_1 \right)^2 + \varepsilon_1\check{x}_2^2 \left(\frac{\partial\alpha_1}{\partial\check{x}_1} \right)^2 \lambda_1\phi_1^2(\check{x}_1) + \varepsilon_1\check{x}_2^2\lambda_2\phi_2(\check{x}_2) + \frac{3}{4\varepsilon_1}\end{aligned}\quad (23)$$

with $\lambda_2 = \lambda_M^2\lambda_{2,1}$.

Design the virtual control as

$$\alpha_2 = -k_2\check{z}_2 + \frac{\partial\alpha_1}{\partial\check{x}_1}\check{x}_2 + \frac{\partial\alpha_1}{\partial\theta_1}\dot{\theta}_1 + \frac{\partial\alpha_1}{\partial\hat{\lambda}_1}\tau_{1,2} + \mathcal{N}(\theta_2)\Phi_2 \quad (24)$$

with

$$\Phi_2 = \varepsilon_1 \frac{1}{\check{z}_2} (\hat{\lambda}_1\omega_{1,2} + \hat{\lambda}_2\omega_{2,1} + \hat{\lambda}_0\omega_{0,1}), \quad \dot{\theta}_2 = \check{z}_2\Phi_2 \quad (25)$$

$$\tau_{1,2} = \omega_{1,2} + \tau_{1,1}, \quad \omega_{1,2} = \check{z}_2^2 \left(\frac{\partial\alpha_1}{\partial\check{x}_1} \right)^2 \phi_1^2(\check{x}_2) \quad (26)$$

$$\tau_{2,1} = \omega_{2,1} - \sigma\hat{\lambda}_2, \quad \omega_{2,1} = \check{z}_2^2\phi_2^2(\check{x}_2) \quad (27)$$

$$\tau_{0,1} = \omega_{0,1} - \sigma\hat{\lambda}_0, \quad \omega_{0,1} = \check{z}_2^2 \left(\alpha_1 - \frac{\partial\alpha_1}{\partial\check{x}_1} \check{x}_1 \right)^2, \quad (28)$$

where k_2 is positive constant to be designed and $\tau_{1,2}$, $\tau_{2,1}$, and $\tau_{0,1}$ are tuning functions.

The time derivative of V_2 becomes

$$\begin{aligned}\dot{V}_2 &\leq -\bar{k}_1z_1^2 - \bar{k}_2z_2^2 + \frac{1}{2}z_3^2 + \sum_{m=1}^2 (g_m(t)N(\theta_m) + 1) \dot{\theta}_m + \sum_{m=1}^2 \tilde{\lambda}_m(\tau_{m,2-m+1} - \dot{\lambda}_m) \\ &\quad + \sigma\tilde{\lambda}_0(\tau_{0,1} - \dot{\lambda}_0) - z_2\lambda \frac{\partial\alpha_1}{\partial\hat{\lambda}_1} (\dot{\lambda}_1 - \tau_{1,2}) - \sigma \sum_{m=0}^2 \tilde{\lambda}_m\hat{\lambda}_m + 4 \times \frac{1}{4\varepsilon_1}\end{aligned}\quad (29)$$

with $\bar{k}_2 = k_2 - 1$, $g_2(t) = \lambda^2$.

Step i , ($3 \leq i \leq n-1$): According to (10), one has

$$\begin{aligned} z_i &= z_{i+1} + \lambda \alpha_i + f_i(\tilde{x}_i) - \dot{\lambda} \alpha_{i-1} - \lambda \dot{\alpha}_{i-1} \\ &= z_{i+1} + \lambda \alpha_i + f_i(\tilde{x}_i) - \dot{\lambda} \alpha_{i-1} \\ &\quad - \lambda \left(\sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \frac{\lambda \dot{x}_m - \dot{\lambda} x_m}{\lambda^2} + \sum_{m=0}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \hat{\lambda}_m} \dot{\lambda}_m + \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \theta_m} \dot{\theta}_m \right). \end{aligned} \quad (30)$$

Similarly, there exist unknown constant $\lambda_{i,1}$ and known smooth function $\phi_i(\tilde{x}_i)$ such that

$$|f_i(\tilde{x}_i)| = |f_i(\lambda, \tilde{x}_i)| \leq \psi_i(\lambda) \phi_i(\tilde{x}_i) \leq \lambda_{i,1} \phi_i(\tilde{x}_i). \quad (31)$$

Here, set the Lyapunov function as

$$V_i = V_{i-1} + \frac{1}{2} z_i^2 + \frac{1}{2} \tilde{\lambda}_i^2, \quad (32)$$

which implies

$$\begin{aligned} \dot{V}_i &\leq \dot{V}_{i-1} + \frac{1}{2} \dot{z}_i^2 + \frac{1}{2} \dot{z}_{i+1}^2 + z_i \lambda \alpha_i - z_i \lambda \left(\sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \dot{x}_{m+1} + \sum_{m=0}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \hat{\lambda}_m} \dot{\lambda}_m + \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \theta_m} \dot{\theta}_m \right) \\ &\quad + \varepsilon_1 \tilde{z}_i^2 \lambda_i \phi_i^2(\tilde{x}_i) + \varepsilon_1 \tilde{z}_i^2 \lambda_0 \left(\alpha_{i-1} - \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \dot{x}_m \right)^2 + \varepsilon_1 \tilde{z}_i^2 \sum_{m=1}^{i-1} \left(\frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \right)^2 \lambda_m \phi_m^2(\tilde{x}_m) \\ &\quad + (i+1) \times \frac{1}{4\varepsilon_1} + \tilde{\lambda}_i \dot{\lambda}_i \end{aligned} \quad (33)$$

with $\lambda_i = \lambda_M^2 \lambda_{i,1}$.

Then, construct the virtual control law as

$$\begin{aligned} \alpha_i &= -k_i \tilde{z}_i + \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \dot{x}_{m+1} + \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \theta_m} \dot{\theta}_m + \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \hat{\lambda}_m} \tau_{m,i-m+1} + \mathcal{N}(\theta_i) \Phi_i \\ &\quad + \frac{\partial \alpha_{i-1}}{\partial \hat{\lambda}_0} \tau_{0,i-1} + \frac{1}{\tilde{z}_i} \sum_{p=1}^{i-2} \sum_{m=p+1}^{i-1} \tilde{z}_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_p} \omega_{p,i-p+1} + \frac{1}{\tilde{z}_i} \sum_{m=3}^{i-1} \tilde{z}_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_0} \omega_{0,i-1} \end{aligned} \quad (34)$$

with

$$\Phi_i = \varepsilon_1 \frac{1}{\tilde{z}_i} \left(\sum_{m=1}^i \hat{\lambda}_m \omega_{m,i-m+1} + \hat{\lambda}_0 \omega_{0,i-1} \right), \quad \dot{\theta}_i = \tilde{z}_i \Phi_i \quad (35)$$

$$\tau_{i,1} = \omega_{i,1} - \sigma \hat{\lambda}_i, \quad \omega_{i,1} = \tilde{z}_i^2 \phi_i^2(\tilde{x}_i) \quad (36)$$

$$\tau_{m,i-m+1} = \omega_{m,i-m+1} + \tau_{m,i-m}, \quad \omega_{m,i-m+1} = \tilde{z}_i^2 \left(\frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \right)^2 \phi_m^2(\tilde{x}_m) \quad (37)$$

$$m = 1, 2, \dots, i-1$$

$$\tau_{0,i-1} = \omega_{0,i-1} + \tau_{0,i-2}, \quad \omega_{0,i-1} = \tilde{z}_i^2 \left(\alpha_{i-1} - \sum_{m=1}^{i-1} \frac{\partial \alpha_{i-1}}{\partial \tilde{x}_m} \dot{x}_m \right)^2. \quad (38)$$

Substituting (34)-(38) into (33) yields that

$$\begin{aligned} \dot{V}_i \leq & -\sum_{m=1}^i \bar{k}_m z_m^2 + \frac{1}{2} z_{i+1}^2 + \sum_{m=1}^i \tilde{\lambda}_m (\dot{\lambda}_m - \tau_{m,i-m+1}) + \sum_{m=1}^i (g_m(t) \mathcal{N}(\theta_m) + 1) \dot{\theta}_m \\ & + \tilde{\lambda}_0 (\dot{\lambda}_0 - \tau_{0,i-1}) - \lambda \sum_{p=1}^{i-1} \sum_{m=p+1}^i z_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_p} (\dot{\lambda}_p - \tau_{p,i-p+1}) - \lambda \sum_{m=3}^i z_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_0} (\dot{\lambda}_0 - \tau_{0,i-1}) \\ & - \sigma \sum_{m=0}^i \tilde{\lambda}_m \dot{\lambda}_m + \left(\frac{(i+1)(i+2)}{2} - 2 \right) \times \frac{1}{4\epsilon_1} \end{aligned} \quad (39)$$

with $\bar{k}_m = k_m - 1$ and $g_m(t) = \lambda^2$ for $m = 3, \dots, n-1$.

Step n: From (10), we have

$$\begin{aligned} z_n &= u + f_n(\bar{x}_n) - \dot{\lambda} \alpha_{n-1} - \lambda \dot{\alpha}_{n-1} \\ &= u + f_n(\bar{x}_n) - \dot{\lambda} \alpha_{n-1} \\ &\quad - \lambda \left(\sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \frac{\lambda \dot{x}_m - \dot{\lambda} x_m}{\lambda^2} + \sum_{m=0}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \hat{\lambda}_m} \dot{\lambda}_m + \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \theta_m} \dot{\theta}_m \right). \end{aligned} \quad (40)$$

Choose the Lyapunov function as

$$V_n = V_{n-1} + \frac{1}{2} z_n^2 + \frac{1}{2} \tilde{\lambda}_n^2 + \frac{1}{2} \tilde{\lambda}_{n+1}^2, \quad (41)$$

which yields

$$\begin{aligned} \dot{V}_n \leq & \dot{V}_{n-1} + z_n (u - \lambda \alpha_n + \lambda \alpha_n) - z_n \lambda \left(\sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \dot{x}_{m+1} + \sum_{m=0}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \hat{\lambda}_m} \dot{\lambda}_m + \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \theta_m} \dot{\theta}_m \right) \\ & + \epsilon_1 \tilde{z}_n^2 \lambda_n \phi_n^2(\check{x}_n) + \epsilon_1 \tilde{z}_n^2 \lambda_0 \left(\alpha_{n-1} - \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \dot{x}_m \right)^2 + \epsilon_1 \tilde{z}_n^2 \sum_{m=1}^{n-1} \left(\frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \right)^2 \lambda_m \phi_m^2(\check{x}_m) \\ & + (n+1) \times \frac{1}{4\epsilon_1} + \tilde{\lambda}_n \dot{\lambda}_n + \tilde{\lambda}_{n+1} \dot{\lambda}_{n+1} \end{aligned} \quad (42)$$

with $\lambda_n = \lambda_M^2 \lambda_{n,1}$ and $\lambda_{n+1} = \lambda_M^4$.

Before giving the actual control law, we introduce the variable α_n

$$\begin{aligned} \alpha_n &= -k_n \check{z}_n + \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \dot{x}_{m+1} + \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \theta_m} \dot{\theta}_m + \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \hat{\lambda}_m} \tau_{m,n-m+1} + \mathcal{N}(\theta_n) \Phi_n \\ &\quad + \frac{\partial \alpha_{n-1}}{\partial \hat{\lambda}_0} \tau_{0,n-1} + \frac{1}{\check{z}_n} \sum_{p=1}^{n-2} \sum_{m=p+1}^{n-1} \check{z}_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_p} \omega_{p,n-p+1} + \frac{1}{\check{z}_n} \sum_{m=3}^{n-1} \check{z}_m \frac{\partial \alpha_{m-1}}{\partial \hat{\lambda}_0} \omega_{0,n-1} \end{aligned} \quad (43)$$

with

$$\Phi_n = \epsilon_1 \frac{1}{\check{z}_n} \left(\sum_{m=1}^n \hat{\lambda}_m \omega_{m,n-m+1} + \hat{\lambda}_0 \omega_{0,n-1} \right), \quad \dot{\theta}_n = \check{z}_n \Phi_n \quad (44)$$

$$\tau_{n,1} = \omega_{n,1} - \sigma \hat{\lambda}_n, \quad \omega_{n,1} = \check{z}_n^2 \phi_n^2(\check{x}_n) \quad (45)$$

$$\tau_{m,n-m+1} = \omega_{m,n-m+1} + \tau_{m,n-m}, \quad \omega_{m,n-m+1} = \check{z}_n^2 \left(\frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \right)^2 \phi_m^2(\check{x}_m) \quad (46)$$

$$m = 1, 2, \dots, n-1$$

$$\tau_{0,n-1} = \omega_{0,n-1} + \tau_{0,n-2}, \quad \omega_{0,n-1} = \check{z}_n^2 \left(\alpha_{n-1} - \sum_{m=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \check{x}_m} \dot{x}_m \right)^2. \quad (47)$$

Design the adaptive laws of the parameters as

$$\dot{\hat{\lambda}}_0 = \tau_{0,n-1}, \quad \dot{\hat{\lambda}}_m = \tau_{m,n-m+1}, \quad m = 1, 2, \dots, n. \quad (48)$$

Then, $\dot{V}_n$ becomes

$$\dot{V}_n \leq \dot{\bar{V}}_n + z_n(u - \lambda\alpha_n) + \tilde{\lambda}_{n+1}\dot{\hat{\lambda}}_{n+1}, \quad (49)$$

where

$$\begin{aligned} \dot{\bar{V}}_n \leq & -\sum_{m=1}^n \bar{k}_m z_m^2 + \sum_{m=1}^n (g_m(t)\mathcal{N}(\theta_m) + 1)\dot{\theta}_m + \left(\frac{(n+1)(n+2)}{2} - 2 \right) \times \frac{1}{4\epsilon_1} \\ & - \sigma \sum_{m=0}^n \tilde{\lambda}_m \hat{\lambda}_m \end{aligned} \quad (50)$$

with $\bar{k}_n = k_n - 1$ and $g_n(t) = \lambda^2$.

Design the actual control law as

$$u = \mathcal{N}(\theta_{n+1})\Phi_{n+1}, \quad \Phi_{n+1} = \tilde{z}_n \hat{\lambda}_{n+1} \alpha_n^2, \quad \dot{\theta}_{n+1} = \tilde{z}_n \Phi_{n+1} \quad (51)$$

with the parameter adaptive law

$$\dot{\hat{\lambda}}_{n+1} = \tilde{z}_n^2 \alpha_n^2 - \sigma \hat{\lambda}_{n+1}. \quad (52)$$

Then,

$$\dot{V}_n \leq -\sum_{m=1}^n \bar{k}_m z_m^2 + \sum_{m=1}^{n+1} (g_m(t)\mathcal{N}(\theta_m) + 1)\dot{\theta}_m - \sigma \sum_{m=0}^{n+1} \tilde{\lambda}_m \hat{\lambda}_m + \left(\frac{(n+1)(n+2)}{2} - 1 \right) \times \frac{1}{4\epsilon_1} \quad (53)$$

with $g_{n+1}(t) = \lambda$.

Here, define

$$\begin{aligned} C &= \min\{2\bar{k}_1, \dots, 2\bar{k}_n, \sigma\} \\ D &= \sum_{m=0}^{n+1} \frac{1}{2} \sigma \lambda_m^2 + \left(\frac{(n+1)(n+2)}{2} - 1 \right) \times \frac{1}{4\epsilon_1}, \end{aligned} \quad (54)$$

and using the inequality $-\sigma \tilde{\lambda}_m \hat{\lambda}_m \leq -\frac{1}{2} \sigma \tilde{\lambda}_m^2 + \frac{1}{2} \sigma \lambda_m^2$ for $m = 0, 1, \dots, n, n+1$, one has

$$\dot{V}_n \leq -CV_n + D + \sum_{m=1}^{n+1} (g_m(t)\mathcal{N}(\theta_m) + 1)\dot{\theta}_m. \quad (55)$$

Now, the main result can be summarized in the following theorem.

Theorem 1. Consider the nonlinear system (1) under false data injection attack (2) and suppose Assumptions 1 and 2 hold. Then, the controller (51) with adaptive laws (48), (52) and constants $k_1 \geq \frac{1}{2}$, $k_m \geq 1$, ($m = 2, 3, \dots, n$) can guarantee the closed-loop system signals globally bounded.

Proof. According to Lemma 3, we have $V(t)$ and θ_m , ($m = 0, 1, \dots, n, n+1$) are bounded, which implies that $V(t)$, z_m ($m = 0, 1, \dots, n$) and λ_m ($m = 0, 1, \dots, n, n+1$) are bounded. Then, the system states x_m , ($m = 0, 1, \dots, n$) and control input u are also bounded. Thus, all the closed-loop signals are globally bounded. $\square$

In the following, the systems with disturbances are discussed:

$$\begin{aligned} \dot{x}_i &= x_{i+1} + f_i(\tilde{x}_i) + d_i(t) \\ \dot{x}_n &= u + f_n(\tilde{x}_n) + d_n(t), \end{aligned} \quad (56)$$

where the disturbances $d_i(t)$ have known upper bounds $\bar{d}_i$, ($i = 1, 2, \dots, n$). Then, by modifying the stabilizing functions in (16), (24), (34), and (43) as

$$\alpha'_i = \alpha_i - \check{z}_i \bar{d}_i^2, \quad i = 1, 2, \dots, n, \quad (57)$$

then, the corresponding adaptive control laws are obtained. Following the systematic design procedure given in Theorem 1, all the signals can be ensured to be bounded. Due to the space limitation, the details are omitted here.

Remark 4. The same attack models have been studied in other works,¹¹⁻¹³ where only the linear systems are considered. Because of the existence of the unknown nonlinearities, the developed control schemes in the aforementioned works¹¹⁻¹³ are failure to ensure the stability of the systems. In this paper, by using backstepping technique and introducing novel coordinate transformations, the adaptive control laws for the CPSs with unknown nonlinearities are designed, where the boundedness of all the signals can be guaranteed.

Remark 5. If $0 < \lambda(t) < 1$, the considered security problem reduces to the sensor fault-tolerant control problem for nonlinear systems, which has been investigated in other works.³¹⁻³³ In this paper, owing to the existence of the attacks through the sensor networks, the sign of $\lambda(t)$ is unknown and may change before and after the occurrence of the attacks, which implies that the proposed control methods in the aforementioned other works³¹⁻³³ are no longer valid. As pointed out in the work of An and Yang,¹³ from the view of system theory, faults and attacks are fundamentally same. However, since the malicious attackers optimize the attack strategies coordinately, the attacks may be undetectable, while there is no collusion between the faults.

4 | SIMULATION EXAMPLES

Example 1. Here, we consider the nonlinear system adopted from the work of Zhai et al³⁴

$$\begin{aligned} \dot{x}_1 &= x_2 + f_1(x_1) \\ \dot{x}_2 &= u + f_2(\bar{x}_2) \end{aligned} \quad (58)$$

with $f_1(x_1) = 0$ and $f_2(\bar{x}_2) = x_1^2 x_2 \sin(x_2)$

The attack weight is given with $\varpi(t) = -3 - 0.25 \cos(t)$ from $t = 20$, where the sign of $\lambda(t)$ changes before and after the attack. The initial states are $x(0) = [0.1, 0.3]^T$, $\lambda_2(0) = \lambda_3(0) = 0$. The parameters are chosen as $k_1 = k_2 = 3$ and $\sigma = 0.001$. The $\mathcal{N}$ function is chosen as $\mathcal{N}(\theta_m) = e^{\theta_m^2} \sin(\pi \theta_m)$.

According to the adaptive control scheme designed in Section 3, the control laws and parameter adaptation laws are given as follows:

$$\begin{aligned} \alpha_1 &= -k_1 \check{z}_1 \\ \alpha_2 &= -k_2 \check{z}_2 + \frac{\partial \alpha_1}{\partial \check{x}_1} \check{x}_2 + \mathcal{N}(\theta_2) \Phi_2 \\ u &= \mathcal{N}(\theta_3) \Phi_3 \end{aligned} \quad (59)$$

with

$$\begin{aligned} \Phi_2 &= \hat{\lambda}_2 \check{z}_2 \phi_2^2(\check{x}_2), \quad \Phi_3 = \hat{\lambda}_3 \check{z}_2 \alpha_2^2 \\ \dot{\hat{\lambda}}_2 &= \check{z}_2^2 \phi_2^2(\check{x}_2) - \sigma \hat{\lambda}_2, \quad \dot{\hat{\lambda}}_3 = \check{z}_2^2 \alpha_2^2 - \sigma \hat{\lambda}_3 \\ \dot{\theta}_2 &= \check{z}_2 \Phi_2, \quad \dot{\theta}_3 = \check{z}_2 \Phi_3. \end{aligned} \quad (60)$$

The state variables and the control input are plotted in Figures 1 and 2, respectively. In addition, the adaptive parameter estimates are depicted in Figure 3.

Figures 1 and 2 show the boundedness of the signals of the closed-loop system, which demonstrate the validity of the algorithm. From Figure 3, it can be seen that $\hat{\lambda}_3$ will change to accommodate the attack when it occurs, which causes the sharp jump in the control input.

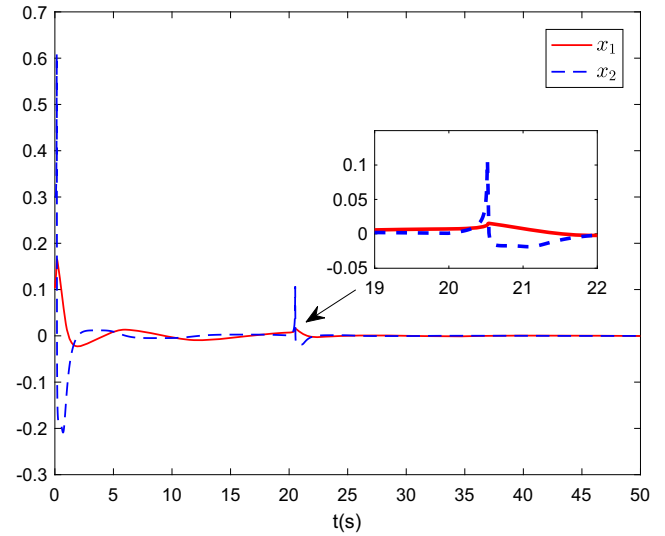


FIGURE 1 The system states [Colour figure can be viewed at wileyonlinelibrary.com]

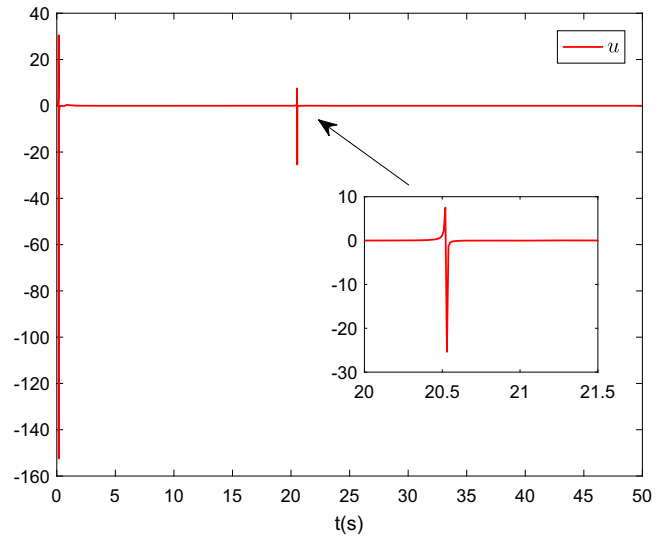


FIGURE 2 The control input [Colour figure can be viewed at wileyonlinelibrary.com]

Example 2. Here, a robotic manipulator driven by a direct current motor borrowed from the work of Dawson et al³⁵ is considered, which can be expressed by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 - \frac{N}{M} \sin x_1 - \frac{B}{M} x_2 \\ \dot{x}_3 &= \frac{1}{LM} u - \frac{K_B}{LM} x_2 - \frac{R}{LM} x_3 \\ y &= x_1\end{aligned}\tag{61}$$

with

$$\begin{aligned}M &= \frac{J}{K_r} + \frac{mL_0^2}{3K_r} + \frac{M_0L_0^2}{K_r} + \frac{2M_0R_0^2}{5K_r} \\ N &= \frac{mL_0G}{2K_r} + \frac{M_0L_0G}{K_r}, \quad B = \frac{B_0}{K_r},\end{aligned}$$

where $x_1 = q$, $x_2 = \dot{q}$, $x_3 = I/M$, $u = V_e$. The meanings of q , $\dot{q}$, I , V_e , R , L , K_B , J , m , L_0 , M_0 , R_0 , G , and K_r can be found in the work of Ren et al.³⁶ Moreover, they are chosen as $J = 1.625 \times 10^{-3} \text{ Kg m}^2$, $m = 0.506 \text{ Kg}$, $M_0 = 0.434 \text{ Kg}$,

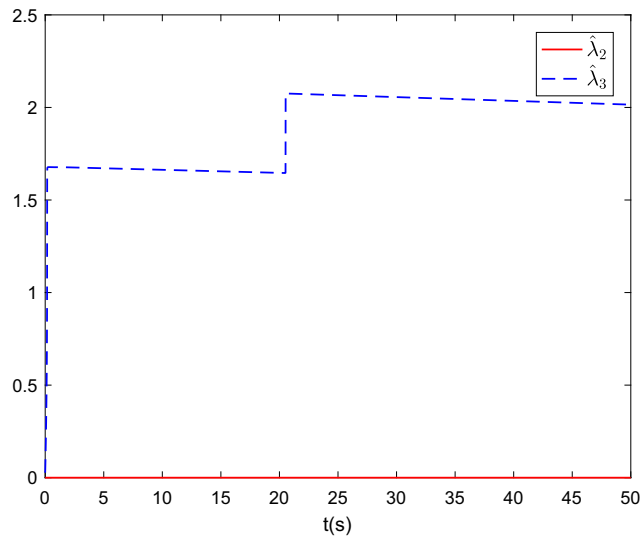


FIGURE 3 The adaptive parameter estimates [Colour figure can be viewed at wileyonlinelibrary.com]

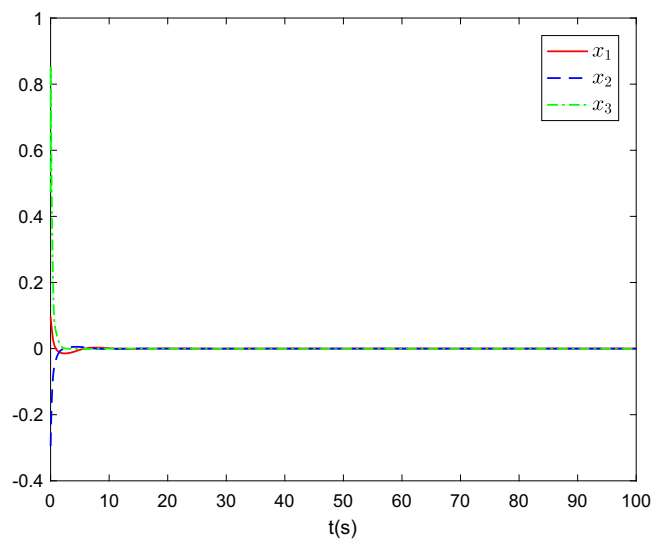


FIGURE 4 The system states [Colour figure can be viewed at wileyonlinelibrary.com]

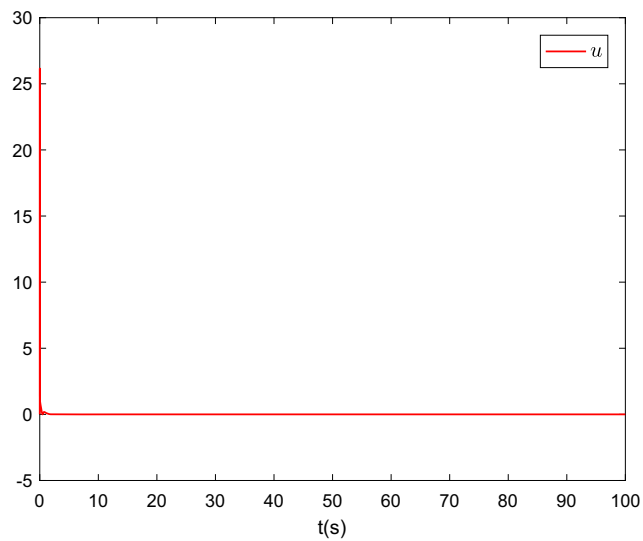


FIGURE 5 The control input [Colour figure can be viewed at wileyonlinelibrary.com]

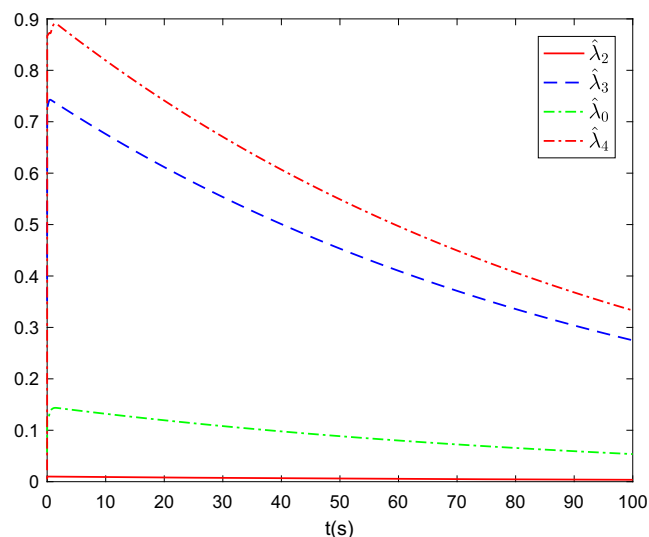


FIGURE 6 The adaptive parameter estimates [Colour figure can be viewed at wileyonlinelibrary.com]

$L_0 = 0.305\text{m}$, $R_0 = 0.023\text{m}$, $B_0 = 16.25 \times 10^{-3}\text{Nms/rad}$, $L = 25.0 \times 10^{-3}\text{H}$, $R = 5.0\Omega$, $K_r = K_B = 0.90\text{Nm/A}$, and $G = 9.8\text{m/s}^2$.

The time-varying sensor attack weight is given with $\varpi(t) = -2 - 1.25 \cos(t)$. The initial states are chosen as $x(0) = [0.1, -0.3, 0.5]^T$, $\lambda_2(0) = \lambda_3(0) = \lambda_0(0) = \lambda_{3,1}(0) = 0$. The parameters are set as $k_1 = k_2 = k_3 = 3$ and $\sigma = 0.01$. The results are depicted in Figures 4 and 5. Furthermore, the adaptive parameter estimates are plotted in Figure 6.

From Figures 4 to 6, it is seen that the control objective is achieved by using the presented control strategy.

5 | CONCLUSIONS

This paper considers the adaptive control problem for a class of nonlinear cyber-physical systems with false data injection attacks through sensor networks. To mitigate the effects caused by the attacks, a novel coordinate transformation is developed in the backstepping control design. Moreover, to deal with the multiple unknown time-varying state feedback gains caused by the sensor attacks, the new types of Nussbaum functions are introduced. By using Lyapunov stability theory, the proposed adaptive controller can guarantee all the closed-loop system signals are globally bounded. Finally, the simulation results are given to show the effectiveness of the proposed method.

ACKNOWLEDGEMENTS

This work was supported in part by the Funds of the National Natural Science Foundation of China under grants 61621004 and 61420106016 and the Research Fund of State Key Laboratory of Synthetical Automation for Process Industries under grant 2018ZCX03.

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How to cite this article: Ren X-X, Yang G-H. Adaptive control for nonlinear cyber-physical systems under false data injection attacks through sensor networks. *Int J Robust Nonlinear Control.* 2020;30:65–79. <https://doi.org/10.1002/rnc.4749>

APPENDIX

THE PROOF OF LEMMA 3

Before giving the proof of Lemma 3, Lemma 4 is needed.

Lemma 4 (See the work of Chen and Huang²⁹). For $\mathcal{N}$ function $\mathcal{N}(s)$ and constants a, b , if $ab > 0$, $\hat{\mathcal{N}}(s) = a\mathcal{N}^+(s) + b\mathcal{N}^-(s)$ is also a class $\mathcal{N}$ function.

Here, the proof of Lemma 3 is given.

Proof. Let $\hat{\mathcal{N}}_m(s) = g_m^+\mathcal{N}^+(s) + g_m^-\mathcal{N}^-(s)$, for $g_m^+g_m^- > 0$, by using Lemma 4, $\hat{\mathcal{N}}(s) \in \mathcal{N}$. For all $\tau \geq 0$,

$$\begin{aligned} g_m(\tau)\mathcal{N}(\theta_m) &= g_m(\tau)\mathcal{N}^+(\theta_m) + g_m(\tau)\mathcal{N}^-(\theta_m) \\ &\leq g_m^+\mathcal{N}^+(\theta_m) + g_m^-\mathcal{N}^-(\theta_m) = \hat{\mathcal{N}}(\theta_m). \end{aligned} \quad (\text{A1})$$

Multiplying (7) by e^{Ct} produces

$$\frac{d(V(t)e^{Ct})}{dt} \leq De^{Ct} + e^{Ct} \sum_{m=1}^n (g_m(t)N(\theta_m) + 1) \dot{\theta}_m. \quad (\text{A2})$$

Then, integrating (A2) over $[0, t]$ generates

$$V(t) \leq V(0)e^{-Ct} + \frac{D}{C}(1 - e^{-Ct}) + e^{-Ct} \int_0^t \sum_{m=1}^n (g_m(\tau)N(\theta_m(\tau)) + 1) \dot{\theta}_m(\tau)e^{C\tau} d\tau. \quad (\text{A3})$$

Since $0 < e^{-C(t-\tau)} \leq 1$, we have

$$\begin{aligned} 0 \leq V(t) &\leq e^{-Ct} \sum_{m=1}^n \int_0^t (g_m(\tau)\mathcal{N}(\theta_m(\tau)) + 1) \dot{\theta}_m(\tau)e^{C\tau} d\tau + c_1 \\ &= e^{-Ct} \sum_{m=1}^n \int_0^t g_m(\tau)\mathcal{N}(\theta_m(\tau))\dot{\theta}_m(\tau)e^{C\tau} d\tau + e^{-Ct} \sum_{m=1}^n \int_0^t \dot{\theta}_m(\tau)e^{C\tau} d\tau + c_1 \\ &\leq \sum_{m=1}^n \int_{\theta_m(0)}^{\theta_m(t)} |\hat{\mathcal{N}}(s)| ds + \sum_{m=1}^n \int_0^t \dot{\theta}_m(\tau) d\tau + c_1 \\ &= \sum_{m=1}^n \int_0^{\theta_m(t)} |\hat{\mathcal{N}}(s)| ds - \sum_{m=1}^n \int_0^{\theta_m(0)} |\hat{\mathcal{N}}(s)| ds + \sum_{m=1}^n (\theta_m(t) - \theta_m(0)) + c_1 \end{aligned} \quad (\text{A4})$$

where $c_1 = V(0) + \frac{D}{C}$.

In the following, we will show the boundedness of $\theta_1(t), \dots, \theta_n(t)$. From (8), we know that the lower bounds of $\theta_m(t)$ are the initial values $\theta_m(0)$. Thus, if $\theta_m(t)$ has upper bounds, then $\theta_m(t)$ is bounded.

Define $\theta_{\max}(t) = \{\theta_1(t), \dots, \theta_n(t)\}$. If $\theta_{\max}(t)$ has the upper bound, $\theta_1(t), \dots, \theta_n(t)$ must have upper bounds. Set $c_2 = \sum_{m=1}^n \int_0^{\theta_m(0)} |\hat{\mathcal{N}}(s)| ds + \sum_{m=1}^n \theta_m(0) - c_1$. One has

$$\begin{aligned} 0 \leq V(t) &\leq \sum_{m=1}^n \int_0^{\theta_m(t)} |\hat{\mathcal{N}}(s)| ds + \sum_{m=1}^n \theta_m(t) - c_2 \\ &\leq n \int_0^{\theta_{\max}(t)} |\hat{\mathcal{N}}(s)| ds + n\theta_{\max}(t) - c_2, \end{aligned} \quad (\text{A5})$$

which implies that

$$\theta_{\max}(t) + \int_0^{\theta_{\max}(t)} |\hat{\mathcal{N}}(s)| ds \geq \frac{c_2}{n}. \quad (\text{A6})$$

Case 1: Consider the case that $\hat{\mathcal{N}}(s) \geq 0$, (A6) becomes

$$\theta_{\max}(t) + \int_0^{\theta_{\max}(t)} \hat{\mathcal{N}}(s) ds \geq \frac{c_2}{n}. \quad (\text{A7})$$

As $\hat{\mathcal{N}}(s)$ is a Nussbaum function, there exists $\theta_{\max}^{*1} > 1$ such that

$$\frac{1}{\theta_{\max}^{*1}} \int_0^{\theta_{\max}^{*1}} \hat{\mathcal{N}}(s) ds < -\frac{|c_2|}{n} - 1. \quad (\text{A8})$$

If $\theta_{\max}(t)$ is not bounded, there exist $t_1^* > 0$ such that $\theta_{\max}(t_1^*) = \theta_{\max}^{*1}$, and hence,

$$\frac{1}{\theta_{\max}(t_1^*)} \int_0^{\theta_{\max}(t_1^*)} \hat{\mathcal{N}}(s) ds < -\frac{|c_2|}{n} - 1 < \frac{c_2}{n\theta_{\max}(t_1^*)} - 1. \quad (\text{A9})$$

As a result,

$$\int_0^{\theta_{\max}(t_1^*)} \hat{\mathcal{N}}(s) ds + \theta_{\max}(t_1^*) < \frac{c_2}{n}, \quad (\text{A10})$$

which contradicts (A7).

Case 2: Consider the case that $\hat{\mathcal{N}}(s) < 0$, (A6) becomes

$$\int_0^{\theta_{\max}(t)} \hat{\mathcal{N}}(s) ds - \theta_{\max}(t) \leq \frac{-c_2}{n}. \quad (\text{A11})$$

As $\hat{\mathcal{N}}(s)$ is a Nussbaum function, there exists $\theta_{\max}^{*2} > 1$ such that

$$\frac{1}{\theta_{\max}^{*2}} \int_0^{\theta_{\max}^{*2}} \hat{\mathcal{N}}(s) ds > \frac{|c_2|}{n} + 1. \quad (\text{A12})$$

If $\theta_{\max}(t)$ is not bounded, there exist $t_2^* > 0$ such that $\theta_{\max}(t_2^*) = \theta_{\max}^{*2}$, and hence,

$$\frac{1}{\theta_{\max}(t_2^*)} \int_0^{\theta_{\max}(t_2^*)} \hat{\mathcal{N}}(s) ds > \frac{|c_2|}{n} + 1 > \frac{-c_2}{n\theta_{\max}(t_2^*)} + 1. \quad (\text{A13})$$

As a result,

$$\int_0^{\theta_{\max}(t_2^*)} \hat{\mathcal{N}}(s) ds - \theta_{\max}(t_2^*) > \frac{-c_2}{n}, \quad (\text{A14})$$

which contradicts (A11).

Combining cases 1 and 2, we can obtain that $\theta_{\max}(t)$ has the upper bound, which implies that θ_m has the upper bounds for $m = 1, 2, \dots, n$. As a result, $V(t)$ must be bounded, which completes the proof. $\square$